

AYUSH KUMAR SINGH

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EDUCATION

ABES Institute of Technology (AKTU)

Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence)

2024 – 2028 | Ghaziabad, India

Current SGPA: 7.13 / 10.0

Sampoorna Kendra Vidyalaya (CBSE)

Senior Secondary (Class XII): **72.0%** | Secondary (Class X): **84.0%**

Dibrugarh, Assam, India

TECHNICAL SKILLS

Frontend & Web: HTML5, CSS3, JavaScript (ES6+), Streamlit, Responsive UI Design

Backend & REST APIs: FastAPI, Python, Asynchronous Programming, RESTful Architecture, Pydantic V2, Uvicorn

Computer Vision & AI: OpenCV, YOLO11, Object Detection, Roboflow (Annotation & Augmentation), NumPy, Pandas, Scikit-Learn

Agentic Workflows & LLMs: LangGraph, LangChain, Multi-Agent Systems, Local LLMs (Ollama, Qwen 2.5), Moondream, Prompt Engineering

Databases & Cloud: PostgreSQL, MongoDB, MySQL Workbench, SQLAlchemy, Schema Normalization, Google Cloud Security

DevOps, Hardware & Tools: NVIDIA DGX A100 / Kubeflow, Docker, Git, GitHub, Linux/Bash, VS Code, Cursor

WORK EXPERIENCE

Global Infoventures (GI Infoventures)

AI & High-Performance Computing (HPC) Trainee

Ghaziabad, India

August 2026 – Present

Computer Vision | Kubeflow | NVIDIA DGX A100 | Object Detection | Roboflow | Docker

- Configured and managed ML workflows on enterprise **NVIDIA DGX A100** infrastructure via Kubeflow notebook servers.
- Engineered computer vision datasets via **Roboflow** for annotation, preprocessing, and augmentation to train detection models.
- Implemented real-time inference workflows using **OpenCV**, benchmarking performance across CPU and MIG GPU slices.
- Containerized training workloads and runtime dependencies using **Docker** for isolated, reproducible execution.

KB QUBE

Machine Learning Intern

Dibrugarh, Assam, India

1 Month

Python | Machine Learning Algorithms | Data Preprocessing | Scikit-Learn | Predictive Modeling

- Completed practical ML assignments focused on supervised/unsupervised algorithms and structured dataset preprocessing.
- Built automated data cleaning, missing-value imputation, and feature scaling pipelines using **Python** and **Pandas**.
- Trained and validated predictive classification models using **Scikit-Learn**, evaluating precision, recall, and accuracy.

KEY PROJECTS

Real-Time Edge Traffic Analytics Pipeline [GitHub]

Python | YOLO11 | ByteTrack | OpenCV | ONNX

- Engineered an end-to-end edge CV pipeline detecting, tracking, and classifying vehicles from 1080p surveillance feeds.
- Integrated **YOLO11 Nano** with **ONNX Runtime** and multi-threaded stream ingestion, sustaining stable **45+ FPS** inference.
- Implemented **ByteTrack** for persistent tracking and built dual virtual tripwires to calculate vehicle velocity (km/h).
- Automated telemetry extraction logging track IDs, timestamps, speeds, and infraction flags into structured data sinks.

Autonomous Multi-Agent AI Travel Orchestrator [GitHub]

Python | LangGraph | Groq (Llama 3) | PostgreSQL

- Architected a multi-agent system in **LangGraph** featuring 4 agents: Flight, Hotel, Itinerary, and Final Aggregator.
- Integrated live tool calling via **Tavily Search** and travel APIs for dynamic rate discovery, route mapping, and scheduling.
- Configured shared state memory (TravelState) with **PostgreSQL** persistence to preserve context across multi-turn sessions.
- Deployed **Groq LPU** (Llama 3) runtime for ultra-low latency response synthesis, cutting generation delay by 40%.

Sovereign AI Workbench & Industrial Inspection [GitHub]

FastAPI | LangChain | Ollama | Moondream | MongoDB

- Engineered an on-premise industrial AI system leveraging local LLMs and vision models for private technical document analysis.
- Implemented a multi-agent pipeline using **LangChain** and **Ollama** to automate component extraction and risk scoring.
- Deployed multimodal vision models (**Moondream / LLaVA**) to detect mechanical anomalies in industrial diagrams.
- Exposed inference via asynchronous **FastAPI** endpoints with Pydantic payload validation and MongoDB persistence.

CERTIFICATIONS & PROFESSIONAL TRAINING

- AI/ML for Geodata Analytics** – ISRO / IIRS
- Trust & Security with Google Cloud** – Google Cloud
- Full Stack Development MasterClass** – NoviTech R&D
- Foundations of Prompt Engineering** – AWS
- Data Analytics Job Simulation** – Deloitte (Forage)
- AI for Business Professionals** – HP LIFE Foundation